An Index Obstruction for Prime Pell Coordinates

A self-contained partial result for OEIS A292082

Primary author (AI): ChatGPT (OpenAI)

November 14, 2026

Abstract

OEIS A292082 lists primes p for which $(p^2-1)/2$ is a square and records a conjectural finite list. We prove that any such prime is the first coordinate of a positive Pell solution whose index is a power of two. This reduces the possible indices to a sparse subsequence; it does not establish finiteness or identify all prime terms.

Sequence and statement

The OEIS entry A292082 concerns primes p with $p^2 - 2q^2 = 1$ for some integer q. Put

$$x_n + y_n\sqrt{2} = (3 + 2\sqrt{2})^n, \qquad n \geq 1.$$

Every positive integer solution of $x^2 - 2y^2 = 1$ with $x > 1$ occurs uniquely in this sequence (the standard Pell parametrization).

Theorem 1. *If x_n is prime, then n is a power of two. Consequently, every prime in A292082 is among $x_1, x_2, x_4, x_8, \ldots$.*

Proof. Write $\alpha = 3 + 2\sqrt{2}$, $\alpha^{-1} = 3 - 2\sqrt{2}$. Then

$$x_n = \frac{\alpha^n + \alpha^{-n}}{2}.$$

For any positive integers a, b, the Chebyshev polynomial T_b satisfies

$$x_{ab} = T_b(x_a), \qquad T_b\left(\frac{z + z^{-1}}{2}\right) = \frac{z^b + z^{-b}}{2}.$$

When b is odd, $T_b(X)$ is an odd polynomial with integer coefficients. Hence X divides $T_b(X)$ in $\mathbb{Z}[X]$, and $x_a \mid x_{ab}$. If also $b > 1$, then $\alpha^{ab} > \alpha^a > 1$, so $x_{ab} > x_a > 1$. Thus x_{ab} is composite.

If n is not a power of two, write $n = ab$ with $b > 1$ an odd divisor and $a \geq 1$. The preceding argument makes x_n composite. This proves the claim. □

Scope and checks

The first candidates are $x_1 = 3$, $x_2 = 17$, $x_4 = 577$, and $x_8 = 665857$, all listed in OEIS A292082. The theorem does not imply that later x_{2^j} are composite. In particular, it does not prove the OEIS finiteness conjecture. The divisibility principle is classical for Pell/Chebyshev sequences; no originality or priority is claimed. The theorem and its proof are AI-generated and have not received independent human peer review.

References

1. OEIS Foundation, *A292082: Primes p such that $(p^2-1)/2$ is a square*, `https://oeis.org/A292082` (accessed November 14, 2026).

2. OEIS Foundation, *A001541: Pell-related sequence*, `https://oeis.org/A001541` (cross-referenced by A292082).